

CHAPTER 6

The Behavior-Analytic Problem-Solving Model

School psychologists use a problem-solving framework as the basis for all professional activities.

—NATIONAL ASSOCIATION OF SCHOOL PSYCHOLOGISTS (2010a, p. 4)

School psychologists use systematic decision-making to consider the antecedents, consequences, functions, and potential causes of behavioral difficulties that may impede learning or socialization.

—NATIONAL ASSOCIATION OF SCHOOL PSYCHOLOGISTS (2010a, pp. 5–6)

Although a functional analysis begins with relatively isolated relations, an important part of its task is to show how its variables interact.

—B. F. SKINNER (1953, p. 205)

In the second edition of *Conducting School-Based Functional Behavioral Assessments: A Practitioner's Guide* (Steege & Watson, 2009), we described and illustrated the behavior-analytic problem-solving (BAPS) model. Since publication of that book, recent research has expanded our knowledge of behavioral assessment and case conceptualization. A revised BAPS model is described and illustrated in this and subsequent chapters. This chapter includes the *rationale* for the use of the BAPS model, a review of the underlying *conceptual foundations*, descriptions of *essential components* of the model, and *clinical examples* that illustrate the application of the model to address school-based referrals.

RATIONALE FOR THE BAPS MODEL

Within the context of FBA, the BAPS model is used to synthesize assessment results and guide the design of individually tailored, function-based interventions. The BAPS model provides an organizational framework for understanding the individual and environmental variables that contribute to the occurrence of interfering behaviors. It facilitates the identification, description, and examination of the collective influences of multiple variables on interfering behaviors. In short, the BAPS model leads to a conceptual analysis of the variables that occasion, evoke, strengthen, and maintain interfering behaviors.

Rather than using the BAPS model as a “stand-alone” assessment process, we use it following the completion of indirect, descriptive, and/or experimental analysis assessments to organize and summarize the findings. Application of the BAPS model is part of our *hypothesis-testing method of assessment* in which (1) hypotheses are formed; (2) assessment data are collected, analyzed, and synthesized; and (3) hypotheses are confirmed or disconfirmed. If hypotheses are confirmed, then we design and implement function-based interventions; if hypotheses are not confirmed, then we go back to the drawing board (“rinse and repeat”). The BAPS model allows us to examine and refine hypotheses based on an analysis of the *interactive effects* of the variables that “trigger” (i.e., evoke and occasion) and maintain (i.e., reinforce) interfering behaviors.

THE EVOLUTION OF CONCEPTUAL MODELS

Sound behavior-analytic conceptual models and a robust body of literature describing the variables that influence human behavior serve as the foundation for the BAPS model. The following are examples of conceptual models that describe functional relations between behavior, antecedent variables, and consequent events.

The Three-Term Conceptual Model

Bijou, Peterson, and Ault (1968) were among the first researchers to assess systematically the environmental variables that trigger and reinforce interfering behaviors. Their A-B-C (i.e., antecedent–behavior–consequence) method for understanding the function(s) of behavior continues to be used by practitioners. Within this A-B-C model:

- *Antecedents* are stimuli that occur prior to behavior and influence its occurrence,
- *Behavior* includes the appropriate or interfering responses produced by the individual, and
- *Consequences* are events that follow behavior and either increase or decrease its future probability.

Only those stimuli that occur prior to behavior and directly influence its occurrence are functional antecedents. Similarly, only those events that follow behavior and directly influence its future occurrence are functional consequences.

The Four-Term Conceptual Model

Goldfried and Sprafkin (1976) expanded the A-B-C model described by Bijou and colleagues (1968) by including a fourth variable for consideration when assessing and conceptualizing the functions of interfering behaviors: *organism* variables. Their stimulus–organism–response–consequence (SORC) model highlighted the fact that antecedents, consequences, and organism variables interact to influence behavior. Nelson and Hayes (1979) defined *organism variables* as “individual differences produced by past learning and by physiology” (p. 7). These include both temporally proximal and distal variables that influence the man-

ner in which individuals respond to their environments (e.g., a history of reading difficulties, vision impairments, or fatigue).

The Five-Term Conceptual Model

In the second edition of this text (Steege & Watson, 2009), we described a five-term conceptual model that expanded on the A-B-C and SORC models by distinguishing between motivational and discriminative antecedent influences on behavior. The model included the following variables:

- Motivating operations
- Discriminative stimuli
- Individual (organism) variables
- Responses (i.e., interfering behaviors)
- Consequences

Unlike the linear A-B-C model, the five-term conceptual model considered the *dynamic* interaction among variables that influenced behavior. MOs were conceptualized as overarching “super variables” that make (1) discriminative stimuli more salient, (2) individual (organism) variables more relevant, (3) interfering behaviors more probable, and (4) reinforcing consequences more valuable.

The Eight-Term Conceptual Model

We now offer an eight-term conceptual model that incorporates the essential components of the A-B-C, SORC, and five-term conceptual models and *adds* the following variables: context, skill deficits/delays, and parameters of reinforcement. This eight-term conceptual framework is the foundation for the BAPS model. As we discuss this model, it will be helpful to refer to the Behavior-Analytic Problem Solving: Assessment Results recording form (Form 6.1, page 105). At this point, consider the recording form as a “trail map” that guides the input and integration of assessment data.

Structure of the Behavior-Analytic Problem Solving: Assessment Results Recording Form

The first thing we want you to take note of is the overall structure of the Behavior-Analytic Problem Solving: Assessment Results recording form. The form facilitates a *synthesis of assessment data* and provides a “big-picture view” of the variables that contribute to interfering behavior. This view allows for a visual analysis of the interactive effects of antecedents and consequences and serves as the foundation for a sound *case conceptualization* with respect to interfering behavior. Notice the positioning of the box for *interfering behavior* (about two-thirds of the way down the page). This is your starting point (the “You Are Here” on the trail map) for completing the form. All of the boxes *above* “Interfering Behavior(s)” refer to variables that precede occurrences of interfering behavior and make it *more probable* in the moment. All of the boxes *below* “Interfering Behavior(s)” describe variables that

follow occurrences of interfering behavior and serve to *maintain and strengthen* it. Later in this chapter, we describe a step-by-step process for inputting all FBA assessment data into the Behavior-Analytic Problem Solving: Assessment Results recording form. But before we go there, please take a few minutes to read through the following descriptions and illustrations of the components of the BAPS model.

Note: If you have previously used the BAPS recording form (Steege & Watson, 2009), you will find this revised version to be more “user-friendly.” Revisions to the form were based on feedback from practitioners, parents, and others who found much of the vocabulary in the original version to be too technical and off-putting. For example, in the current revision, we collapsed two categories—Unconditioned Motivating Operations and Conditioned Motivating Operations—into the single category of Motivational “Triggers.” Technically, MOs do not *trigger* behavior; instead, MOs make reinforcing consequences more valuable and interfering behaviors more probable. We decided to use the term *trigger* because it helps consumers understand that these motivational stimuli and events are antecedents to interfering behavior. Likewise, we redefined S^Ds as “sources of reinforcement” to capture how these antecedent stimuli signal the availability of reinforcement.

COMPONENTS OF THE BAPS MODEL

The following sections describe and illustrate the components of the BAPS model, which is comprised of the following variables:

1. Context
2. Personal characteristics
3. Skill delays/deficits
4. Motivational triggers (MOs)
5. Sources of reinforcement (S^Ds)
6. Interfering behavior
7. Reinforcing consequences
8. Parameters of reinforcement

We suggest that you reference a blank copy of the Behavior-Analytic Problem Solving: Assessment Results recording form (Form 6.1) while reading. Each of the boxes in the recording form is a placeholder: a repository for the information gleaned from the FBA. Yet, the BAPS model is also dynamic, and identification of one component of our eight-term conceptualization of interfering behavior often influences and informs our understanding of additional functional variables. For example, identification of deprivation of social attention as an MO informs the hypothesis that social attention is a reinforcing consequence for interfering behavior. Likewise, the hypothesis that an interfering behavior is maintained by negative reinforcement leads us to examine and identify functionally related MOs (i.e., antecedent events characterized by the presentation of aversive stimuli), and identification of a relevant personal characteristic prompts examination of other associated variables such as skill deficits and delays.

The following subsections include a definition for each component of the BAPS model, followed by a description of the process for identifying the relevant variables, a rationale for including the variables in the conceptualization of interfering behavior, and clinical examples illustrating the relevant concepts.

Interfering Behavior

Definition

Interfering behaviors are behaviors that interfere with the student's progress. These include a variety of responses that are disruptive to the learning of the student and/or others.

Process

Identify and operationally define each response class of interfering behaviors.

Rationale

FBA seeks to understand why students engage in interfering behaviors so that effective interventions can be developed. Therefore, identifying and defining the behaviors targeted for assessment and intervention is an essential first step in the BAPS model.

When defining interfering behaviors, it is important to be specific. Operational definitions provide team members with a common understanding of the "problem" and increase the likelihood of accurate and reliable data collection. It also is important to consider the concept of response classes. Multiple topographies of interfering behavior (e.g., hitting, pinching, and kicking) may form a single response class, which means that each response in the set may be controlled by the same functional antecedents and consequences. In these situations, the conceptualization of the interfering behavior may be completed using a single BAPS recording form. In other situations, different topographies of interfering behavior (e.g., aggression, self-injury, opposition, and tantrum) may be controlled by different antecedents and consequences and/or associated with different contexts, personal characteristics, and skill deficits. Accordingly, each interfering behavior may need to be analyzed using multiple BAPS recording forms.

Example

Steve is a middle school student who exhibits inappropriate verbal behaviors, including swearing at teachers and swearing at peers. Results of the FBA revealed that swearing at teachers resulted in negative reinforcement (avoidance of and escape from tasks), whereas swearing at peers resulted in positive reinforcement (social attention/reactions). Assessment data also revealed that these behaviors occurred in different contexts, were associated with different personal characteristics and skill deficits, and were evoked by different antecedent events. In this case, synthesizing results for all inappropriate verbal behaviors was an inappropriately global approach, and using separate BAPS recording forms to conceptual-

ize “swearing at teachers” and “swearing at peers” proved to be a more effective strategy. Separate analyses were essential for designing different function-based support plans.

Additional Considerations

All information you input into the BAPs recording form (e.g., context, personal characteristics, delays/deficits) must be *relevant to the identified interfering behavior*. Inputting *irrelevant information* compromises the integrity of the conceptual analysis. As you use this form, it is critical to remember that identification of each class of interfering behaviors is the starting point, and only FBA data directly relevant to each class of interfering behaviors should be added to your conceptual map.

Context

Definition

Context refers to the setting or environment in which the interfering behavior occurs. Context may include descriptions of the physical locations, social circumstances, or types of activities associated with interfering behaviors.

Process

Specify the contexts in which interfering behavior is most probable. The normative behaviors expected within the context also may be identified.

Rationale

Interfering behaviors occur within a larger context, and behaviors that occur in different contexts may have unique sets of controlling variables. Knowing the context of interfering behaviors allows us to anticipate the possibility of interfering behavior and understand the normative behavioral expectations. Identifying the contexts associated with interfering behavior provides information about when and where behavioral interventions need to be implemented and informs the selection of target replacement behaviors and treatment strategies (e.g., antecedent modifications).

Example 1

A high school student exhibited interfering behavior (i.e., acting “goofy” and making irrelevant comments) *during language arts class*. The interfering behavior was reinforced by social attention from classmates (socially mediated positive reinforcement). In the context of language arts class, students are expected to participate in group discussion, read orally, read silently, attend to instruction, and attend to peers’ contributions. Therefore, intervention during language arts class focused on increasing appropriate behavior (i.e., relevant comments, sustained oral/silent reading, attention to task) and reducing interfering behaviors.

Example 2

A student displayed inappropriate social behavior with peers *during unstructured classroom situations*. Reducing the number and duration of unstructured situations and providing planned activities during these situations reduced the opportunity for the student to engage in interfering behaviors.

Personal Characteristics

Definition

Personal characteristics include the student's individual traits, sensitivities, beliefs, and/or values that contribute to interfering behavior. (*Note:* It is important to recognize that personal characteristics do not *cause* interfering behavior; rather, they make it challenging for the student to navigate and successfully participate in the context and increase the probability of interfering behaviors.)

Process

Describe the personal characteristics that contribute to interfering behavior. Consider factors such as the student's characteristic patterns of feeling, thinking, and acting; cultural background; and biological/medical conditions and diagnoses.

Rationale

Personal characteristics are the variables that students “bring to the table” as a result of their unique biology and learning history. Identification of relevant personal characteristics helps us pinpoint specific skill delays/deficits and understand why certain events operate as MOs or reinforcing consequences for interfering behavior. For example, due to biology and learning history, a student may be particularly sensitive to some environmental events (e.g., sudden loud noises, large crowds, or the behavior of peers), display characteristic behavioral traits and patterns of thinking (e.g., rigidity, resistance to change, impulsivity, low frustration tolerance, and self-deprecating or irrational thoughts), and/or adhere to specific values (e.g., personal or cultural “codes”).

Example 1

Harold is a high school student diagnosed with autism spectrum disorder (ASD) who is hypersensitive to abrupt, loud noises (e.g., the school bell and peer laughter or screaming). FBA results indicated that auditory stimuli were antecedents to Harold's engagement in interfering behaviors such as repeatedly yelling “Stop!” Using the BAPS model to understand Harold's interfering behavior, we identified *sensitivity to abrupt, loud noises* as a relevant *personal characteristic*.

Now, what about the abrupt loud noises? In Harold's case, these variables operated as MOs. That is, the presentation of auditory stimuli such as the school bell and sudden peer

laughter or screaming increased the value of negative reinforcement in the form of escape from aversive stimuli.

And how was reinforcement in the form of escape delivered as a consequence for interfering behavior? Well, the school bell rings for only a few seconds, and Harold had associated yelling “Stop!” with the cessation of the bell ringing. In other words, the school bell stopped ringing shortly after he yelled. The two events were not *causally* related; the yelling did not directly result in the termination of the bell ringing. Nonetheless, yelling behavior was reinforced by the cessation of the bell. It was a form of “superstitious behavior.” On the other hand, yelling in response to peer laughter and screaming was maintained by socially mediated negative reinforcement. When Harold yelled “Stop!” his teacher typically escorted him to a quiet adjacent classroom.

So how did this information inform intervention? Antecedent modification is a strategy for reducing interfering behavior by altering or eliminating the relevant MOs. In this case, it was not practical to alter or eliminate the school bell. Moreover, the laughter and screaming of classmates was not predictable or controllable, and there were many naturally occurring situations in which that behavior was perfectly acceptable (e.g., during basketball games, school assemblies, and teen idol visits to the school). The team therefore decided to implement a self-management program. The school psychologist used prompting and reinforcement strategies to teach Harold specific coping skills as a replacement for yelling. He participated in daily role-play sessions to practice these skills in response to simulated triggers (e.g., the school psychologist screaming like a peer). He also carried a written script of the coping skills routine, self-monitored and self-reinforced engagement in coping skills, and wore a wristwatch that beeped 1 minute before the school bell rang to prompt coping skill practice. This intervention package resulted in an immediate reduction and eventual elimination of Harold’s inappropriate yelling behavior.

Example 2

Jaymee is a fifth-grade student with delays in reading fluency and comprehension who makes frequent errors (e.g., mispronunciations, word substitutions, and word omissions) while reading orally. Jaymee was referred for an FBA of verbal and nonverbal oppositional–defiant behaviors that typically occurred during small-group reading instruction in a resource classroom. The FBA included (1) behavior-analytic interviews with Jaymee and her special education teacher and (2) direct observations during small-group reading instruction. The special education teacher reported that Jaymee was sensitive to corrective feedback during reading and concerned about peer appraisal. When interviewed, Jaymee confirmed these observations and displayed self-deprecating and irrational patterns of thinking. For example, she commented: “Everyone thinks I’m stupid,” “My teachers think I’m the dumbest kid in the class,” and “I completely suck at reading.” Given these personal characteristics, interfering behaviors became probable in the context of small-group reading instruction when Jaymee was expected to read aloud. In this case, both *sensitivity to corrective feedback* and *irrational beliefs* were documented as relevant *personal characteristics* on the BAPS recording form.

Skill Delays/Deficits

Definition

Skill delays and deficits refer to appropriate behaviors that are absent or limited and thereby impair the student's ability to cope effectively with personal characteristics and/or respond successfully in specified contexts.

Process

Specify the communication, academic, executive, adaptive living, social, and/or self-management skills that, when absent or limited, contribute to interfering behavior.

Rationale

In our experience, the vast majority of interfering behaviors displayed by students are directly related to skill delays or behavioral deficits. In fact, in most cases, we are able to reframe interfering behaviors as behavioral deficits. For example, interfering behavior maintained by negative reinforcement (escape from tasks) may be reframed as a deficit in appropriate communication skills (e.g., requesting a break or asking for help). Similarly, aggressive behavior between classmates may be reframed in terms of conflict resolution skill deficits, and stereotypy maintained by automatic reinforcement (arousal induction/sensory stimulation) may be reframed as a deficit in independent leisure skills. Each behavior deficit thus links to a skill that may be taught and reinforced as a replacement for interfering behavior. This is why identifying the contexts in which interfering behaviors occur helps the team identify logical replacement behaviors. We ask two questions: (1) What are the normative behaviors and academic/social expectations within the relevant context? and (2) Does the student have the requisite skills to engage in these normative and expected behaviors? Then we advocate for the "fair-pair" model of intervention. That is, it is only "fair" to "pair" an appropriate behavior to increase with each interfering behavior we plan to reduce.

Example 1

Odessa frequently exhibited disruptive behaviors (e.g., verbal opposition and property destruction) when verbally directed to participate in cooperative learning activities. Odessa has a strong preference for completing assignments individually and explained that she likes to complete work "on my own terms, in my own sweet time, and with my own flair." She also has a history of arguing with classmates and engaging in disruptive behavior when "she doesn't get her own way." The FBA resulted in identification of several behavioral deficits, including *inadequate conflict resolution, collaboration, and negotiation skills*. The school counselor and school psychologist collaborated in offering the Strong Kids program (Merrell, Juskelis, Tran, & Buchanan, 2008) to address these behavioral deficits. Engaging Odessa in the Strong Kids curriculum within a small group allowed for modeling, practice, and reinforcement of prosocial replacement behaviors.

Example 2

Consider the previous case of Jaymee. *Deficits in reading accuracy, fluency, and comprehension skills* clearly contributed to occurrences of interfering behavior. Additionally, Jaymee's irrational beliefs revealed *deficits in the skill of disputing automatic thoughts and using positive self-talk*.

Executive Skills

Table 5.1 in Chapter 5 (pp. 55–56) links common referral concerns with specific executive skill weaknesses. We find this to be a useful guide as we reframe personal characteristics as skill delays/deficits that can be directly targeted for intervention.

Motivational Triggers (MOs)

Definition

Motivating operations include antecedent stimuli and events that temporarily (1) alter the effectiveness of reinforcing consequences and (2) increase the likelihood of interfering behaviors maintained by those currently valuable consequences.

Process

Specify antecedent stimuli and conditions that *evoke* interfering behavior. Consider conditions of deprivation (e.g., reduced or restricted access to preferred edibles, tangibles, activities, and social interactions) and the presentation of aversive stimuli or events (e.g., pain, unpleasant thoughts and emotions, challenging academic tasks, and undesirable social interactions).

Rationale

Interfering behaviors are often triggered by the deprivation of reinforcing stimuli or the presentation of aversive stimuli. These motivational variables are in constant flux, which explains why student behavior can be so variable! Identification of the MOs that evoke interfering behavior is essential for developing antecedent-based interventions that reduce the probability of interfering behaviors by preventing conditions of deprivation and altering the aversive aspects of the student's environment.

Example 1

A student has a history of food-stealing behavior. Deprivation of food temporarily (1) increases the value of food as a reinforcer and (2) increases the probability of food-seeking behavior. When the student was placed on a *low-calorie diet* (an MO), food became extremely valuable and he engaged in high rates of food-stealing behavior. When the student was permitted free access to high-quality snacks, the motivation to access food and the likelihood of food-stealing behavior were reduced.

Example 2

A high school junior has a history of sleep apnea and chronic migraine headaches. Due to sleep apnea, the student experiences *sleep deprivation* and *painful stimulation caused by migraine headaches* (both of which operate as MOs). Both of these antecedent variables increase the value of escaping from lecture-type instructional sessions (negative reinforcement) and increase the probability of off-task behaviors, such as doodling, laying her head on the desk, and sleeping in class. Additionally, sleep deprivation and headache pain may decrease the value of reinforcers that typically maintain on-task behavior (e.g., teacher praise and earned access to break-time rewards).

Sources of Reinforcement (S^D s)

Definition

Discriminative stimuli (S^D s) are antecedent stimuli that are correlated with the availability of reinforcement and set the occasion for behavior.

Process

Specify the S^D s that signal the availability of reinforcement and therefore trigger interfering behavior. Consider potential sources of reinforcement, including the presence of reinforcing items and people correlated with the delivery of reinforcement for interfering behaviors.

Rationale

When we conduct an FBA, we conceptualize the S^D as the *source of reinforcement*. When interfering behaviors are maintained by individually mediated reinforcement, the items, location, and situations that the student wants to access or avoid typically function as S^D s. In contrast, when behaviors are maintained by socially mediated reinforcement, the persons delivering the reinforcement operate as S^D s. Interfering behaviors are more likely to occur when relevant S^D s are present; therefore, identification of S^D s informs antecedent intervention strategies that involve removing or modifying these triggers.

Example 1 (Individually Mediated Reinforcement)

Steve is a senior in high school. As part of a transition plan to prepare him for employment at a local apple orchard, Steve performs several school “jobs” including picking, sorting, cleaning, and bagging McIntosh apples. Steve was referred for an FBA to address compulsive cell-phone checking that was interfering with his job performance. The job coach reported that Steve frequently stops working, removes the phone from his pants pocket, and checks for messages. On a day when Steve forgot his cell phone at home, Steve’s productivity was 75% greater than usual. In this example, the *cell phone* is an S^D for compulsive cell-phone checking behavior. The cell phone signals the availability of potential reinforcing messages, and it was readily available in Steve’s pants pocket. Directing Steve to store the phone in a

locker that was accessible only during scheduled break times reduced work stoppages and increased productivity.

Example 2 (Socially Mediated Reinforcement)

Jarry is a preschool student who displays disruptive behaviors during morning circle time when students recite the days of the week, check the weather, and sing the alphabet. Sometimes Teacher A leads the group activity, and sometimes Teacher B leads the activity. During circle time observations, the evaluator noticed a pattern, namely that Jarry's disruptive behavior occurred more frequently and for longer durations when Teacher A led the activity than when Teacher B led the activity. Additional observations revealed that Teacher A typically responded to disruptive behavior by asking Jarry to sit next to her and then providing tactile supports (e.g., placing her hand on his leg or her arm on his shoulder). In contrast, when Teacher B was directing the activity and Jarry engaged in disruptive behavior, she typically praised other students for "good listening behaviors." In short, Teacher A provided social attention contingent on disruptive behaviors, whereas Teacher B ignored disruptive behaviors and differentially reinforced appropriate behaviors. In this example, *Teacher A* functions as an S^D , signaling the availability of socially mediated positive reinforcement and occasioning disruptive behavior. Technically, Teacher B also serves as a type of discriminative stimulus (an S -delta [S^Δ]) that signals the *unavailability* of reinforcement for disruptive behavior.

"Who Ya Gonna Call?"

Do any of you remember a time in your life when you were "dead broke" or at least really strapped for cash? I (Steege) do! As an undergraduate at Iowa State University, I mismanaged my finances and found myself in need of a quick transfer of funds to cover a check I had written to help pay for a keg of beer. (I was counting on my buddies to reimburse me for the deposit and costs, but they were slow in paying up.) So, I called home and pled my case for a short-term loan. In my case, Parent A was an S^D signaling the availability of sympathy and a quick transfer of funds. Parent B was an S^{DP} for social disapproval and an S^Δ signaling the unavailability of reinforcement. So, who ya gonna call? Parent A, of course. Whewww . . . problem solved!

A couple of decades later I got "the call" from each of my kids. Cool, I'm an S^D —signaling the availability of reinforcements—so I got that going for me, which is nice.

Reinforcing Consequences

Definition

Reinforce means to strengthen, and reinforcement is the process of strengthening a behavior. Reinforcing consequences are stimulus changes that occur subsequent to a response and strengthen the *future* probability of similar responses. When interfering behaviors result in access to preferred stimuli, we classify the consequence as positive reinforcement. When interfering behaviors result in postponement or termination of aversive stimuli, we classify the consequence as negative reinforcement. Reinforcing consequences may be delivered by other people (socially mediated) or result directly from the behavior (individually mediated and automatic).

Process

Identify the consequences that follow and reinforce (strengthen) interfering behavior. Classify the consequences as positive (access) or negative (avoidance/escape), and determine whether the consequences are individually mediated, socially mediated, or automatic.

Rationale

Differentiating the specific consequences that reinforce and maintain interfering behaviors is critical to understanding *why* these behaviors occur. Moreover, knowledge of the functional reinforcer(s) for interfering behaviors directly informs intervention: Once we identify a relevant reinforcing consequence, we can withhold it contingent on interfering behavior and provide it contingent on an appropriate replacement behavior.

Remember, reinforcing consequences strengthen the *future* probability of behavior. When conducting an FBA, it is important to distinguish between the *immediate* effect of a consequence and the *future* effect of a consequence. For example, a preschooler, Tiffany, exhibited tantrum behaviors when a classmate, Russ, unexpectedly grabbed her favorite color crayon during a free-time activity. When the teacher assistant intervened and returned the crayon to Tiffany, she immediately stopped crying. From the teacher assistant's perspective, returning the crayon "worked" because the tantrum behavior stopped. However, in the future, Tiffany's tantrum behavior increased markedly, especially when other children used her preferred classroom materials. Tiffany subsequently was referred for an FBA because of increased tantrum behaviors and "problems with sharing." Obviously, from a behavior-analytic perspective, the return of the crayon (and other items) did work . . . it worked to reinforce tantrum behaviors. Given this scenario, what types of strategies may be effective in (1) reducing tantrum behaviors and (2) increasing sharing behaviors?

Example 1

When Erin gossiped about two particular peers at her school, her closest friends provided increased quality attention (e.g., eye contact with widened eyes, enthusiastic reciprocal social interactions, and invitations for sleepovers). Erin is now engaging in high rates of gossip behaviors. In this case, the reinforcing consequence for gossiping was *access to social attention from peers* (socially mediated positive reinforcement).

Example 2

Roxanna is a preschool student who prefers to play alone and often engages in severe tantrum behavior when she is prompted to stop playing and join group activities. The team decided to use a planned ignoring strategy (i.e., ignore Roxanna and engage the other children in the activity) when Roxanna exhibited tantrum behaviors in response to prompts to join the group. This strategy was based on the hypothesis (hope?) that Roxanna would grow tired of playing alone, feel jealous about the attention provided to her classmates, and then (eventually) join the group. Now, whenever Roxanna is asked to participate in group activities, she immediately engages in tantrum behavior. In this example, Roxanna's tantrum behaviors

were reinforced by *avoidance of group activities* (negative reinforcement) and *continued access to solitary play* (positive reinforcement). We call this the “double whammy.”

Example 3

Belynda is a 19-year-old student who displays high rates of pacing and hand wringing (i.e., she paces back and forth while simultaneously clutching and wringing her hands). This behavior typically occurs when Belynda is not engaged in community living, daily living, recreation-leisure, self-help, or vocational tasks. Based on a comprehensive FBA, the school psychologist determined that the behaviors appeared to be maintained by *sensory consequences* (automatic reinforcement).

Parameters of Reinforcement

Definition

Parameters are the dimensions of reinforcement that influence its strengthening effects. Key parameters of reinforcement include schedule, quality, magnitude, and immediacy.

Process

Identify or estimate the following parameters of reinforcement:

- *Schedule*: How frequently is reinforcement delivered contingent on the interfering behavior?
- *Quality*: How valuable or preferred is the reinforcing consequence?
- *Magnitude*: How large is the reinforcer or period of access to the reinforcer?
- *Timing*: How immediate or delayed is the delivery of the reinforcing consequence?

Rationale

To compete with the reinforcing consequences maintaining interfering behaviors, we need to understand *what we are up against*. Describing the schedule, quality, magnitude, and timing of reinforcement provides the information we need to determine the *robustness* of the reinforcing consequence (i.e., the strength of the behavior–consequence relationship). We can then use this information to arrange more robust reinforcing consequences for replacement behaviors and *effectively compete* with the reinforcement that follows interfering behavior.

Examples (Schedules of Reinforcement)

Schedules of reinforcement describe how often and when interfering behaviors produce reinforcement. The following are examples of schedules of reinforcement maintaining interfering behavior.

- Every time a student engages in physical aggression, he is sent home for the day, and he really likes going home! (continuous reinforcement).
- When a student calls out in class, she gets a check mark by her name on the board. After three check marks (i.e., three call-outs), she is required to stay inside with the teacher for recess. For the student, this is a good thing because she gets teased on the playground and loves one-on-one time with her teacher (intermittent reinforcement; fixed ratio 3).
- A student engages in inappropriate joke telling. On average, every third inappropriate joke results in positive reinforcement in the form of peer laughter (intermittent reinforcement, variable ratio 3).
- The classroom teacher wanders the room when students complete independent seatwork. When a student is off-task and the teacher notices, the teacher engages the student in conversation and offers assistance. After varying durations of off-task behavior, the student accesses positive reinforcement in the form of teacher attention (intermittent reinforcement, variable interval).

Examples (Quality of Reinforcement)

Reinforcement varies in terms of quality, and quality is individually determined. I *love* peanut M&Ms (high quality). I *like* Dove dark chocolates (medium quality). And carob morsels, the healthy alternative when making chocolate chip cookies, *will do in a pinch* (low quality). The question here is: To what degree is the reinforcing consequence valued or preferred by the student? For example:

- How valuable is social attention? Is the attention physical or verbal? Is the tone neutral or enthusiastic? Is it from teachers or the cool kids?
- How valuable is escape from a task? Are tasks simply removed or does the student get to take a break in a preferred location with preferred activities?
- How valuable is the edible, tangible, or activity reinforcer? Does the student get access to coloring (low preference) or the iPad (high preference)? Does the student get a stale cup of Folgers coffee or a piping hot caramel latte from Star-Dunkin?

Keep in mind too that the quality of reinforcement is variable and impacted by fluctuating MOs. For example, peanut M&Ms may be my favorite, but, after eating a double pounder of peanut M&Ms, the thought of chocolate may be nauseating.

Examples (Magnitude of Reinforcement)

The magnitude of reinforcement refers to the *amount* of reinforcement. For example:

- 10–20 seconds of social attention
- 8–12 minutes of escape from classroom activities
- 5 minutes of access to the iPad
- 10 peanut M&Ms

Examples (Timing of Reinforcement)

The timing of reinforcement refers to the amount of time that elapses between the behavior and the reinforcing consequence. Timing may be immediate (e.g., 2–5 seconds after the behavior) or delayed (e.g., 1 minute or 1 hour after the behavior). For example, positive reinforcement in the form of individualized attention from the teacher may occur immediately after an occurrence of interfering behavior or at the end of the day during a conference to reflect on prior occurrences of interfering behavior. Generally speaking, the more immediate the delivery of the reinforcing consequence, the stronger the behavior–reinforcer relationship.

Summary

The BAPS model facilitates a conceptualization of interfering behavior as a function of dynamic interactions between multiple environmental variables. This eight-term conceptualization of behavior recognizes that assessment of the school context and the unique characteristics and skills of each student sets the foundation for understanding the manner in which antecedents and consequences directly occasion, evoke, and maintain interfering behaviors. The following is a summary of the functional properties of the variables within the BAPS model that contribute to interfering behaviors.

Context

- Context makes interfering behavior more *predictable*.
- Context makes personal characteristics more *salient*.
- Context makes skill delays/deficits more *pronounced*.

Personal Characteristics and Skill Delays/Deficits

- Individual variables make it more *challenging* for students to be successful within a particular context.

Motivational Triggers (MOs)

- MOs make discriminative stimuli more *relevant* and *effective*.
- MOs make reinforcing consequences more *valuable*.
- MOs make interfering behavior more *probable*.

Sources of Reinforcement (S^Ds)

- S^Ds make interfering behavior more *probable*.

Reinforcing Consequences

- Reinforcing consequences *strengthen* interfering behaviors by increasing the future frequency, duration, and/or intensity of behavior.

Parameters of Reinforcement

- Parameters of reinforcement impact the relative strength (*robustness*) of reinforcing consequences.

USING THE BEHAVIOR-ANALYTIC PROBLEM SOLVING: ASSESSMENT RESULTS RECORDING FORM

We have designed the Behavior-Analytic Problem Solving: Assessment Results recording form to assist with organizing and synthesizing FBA data in a meaningful way. (In the old days, we sat in front of a desk that looked like the chart table of the *Titanic* as she was foundering at sea and attempted to make sense of vast collections of data such as results of adaptive behavior, curriculum-based measures, and preference assessments; pages of interview notes; raw and tabulated data from anecdotal and descriptive observations; and graphs of brief or extended functional analyses. It was a mess. And our FBA reports were long narratives that attempted to organize the results but put even the most avid behavior analyst to sleep.) The Behavior-Analytic Problem Solving: Assessment Results recording form simplifies the interpretation of FBA results by offering a visual format to display the interrelationships between contributing variables (context, personal characteristics, skill delays/deficits), functional antecedents (MOs and S^Ds), reinforcing consequences, and interfering behaviors.

The following is a guide for completing the Behavior-Analytic Problem Solving: Assessment Results recording form. You will notice that we do not recommend completing the form from top to bottom; rather, we recommend a step-by-step process that facilitates a functional conceptualization. Remember, although we may use the BAPS model to guide the FBA process, we use the Behavior-Analytic Problem Solving: Assessment Results recording form *following* the completion of a comprehensive FBA.

Step 1: Interfering Behavior

Identifying and defining the interfering behavior is always the starting point, and the conceptual map branches out from here. We suggest limiting your analysis to one interfering behavior or a cluster of interfering behaviors that are members of the same response class.

Step 2: Context

Identifying the context next sets the stage for the analysis of associated and functional variables. The context helps us focus our attention on the skills needed for successful functioning (which guides us to relevant personal characteristics and skill delays/deficits) and the people, interactions, items, activities, and so forth that may function as antecedents or consequences for interfering behaviors.

Step 3: Reinforcing Consequences

Identifying the reinforcing consequences for interfering behavior provides clues about functional antecedents (MOs and S^Ds). For example, when an interfering behavior results in escape from academics, we know that we need to examine which aspects of the academic task are aversive and thus make escape valuable. In some cases, you may find that a single interfering behavior is reinforced by multiple reinforcing consequences (e.g., negative, positive, and automatic reinforcement).

Step 4: Motivational Triggers (MOs)

Now that you have identified the reinforcing consequences, back up and identify the relevant MOs. The key is to determine whether the interfering behavior is preceded by some level of deprivation of (or restricted access to) preferred stimuli or events or by the presentation of aversive (or nonpreferred) stimuli or events—or by both.

Step 5: Sources of Reinforcement (S^Ds)

After identifying the reinforcing consequences, you are well positioned to pinpoint the sources of reinforcement. In many school-based situations, reinforcing consequences are delivered (mediated) by peers, teachers, or other school staff. In those cases, the person with the history of reinforcing the interfering behavior is typically the functional S^D. However, when interfering behaviors result in direct access to reinforcement (individually mediated or automatic reinforcement), we suggest focusing your attention on specific items, locations, or situations that have been correlated with reinforcement.

Interfering Behavior, Context, Reinforcing Consequences, MOs, and S^Ds

Joey is a 17-year-old who lives at home with his single mother and three younger siblings. Joey's mother reported that he never displays interfering behaviors when she is available to provide him with her undivided attention. However, when she is busy (e.g., cooking dinner) and Joey is seated in his wheelchair in the living room by himself, he often engages in loud vocalizations and hand-biting behavior. When she halts her task to check on Joey and see if he is okay, he typically quiets and stops biting his hand. She explained that this pattern repeats itself several times each evening and that talking in a "soothing tone" seems to work because it "settles Joey down." In this example, we can identify the five components of the BAPS model discussed thus far:

1. *Interfering Behavior*: Loud vocalizations and hand biting
2. *Context*: Sitting in a wheelchair alone in the living room
3. *Reinforcing consequences*: Parental attention (socially mediated positive reinforcement)
4. *Motivational triggers (MOs)*: Diverted attention (deprivation of attention)
5. *Sources of reinforcement (S^Ds)*: Joey's mother

Step 6: Personal Characteristics

Now is the time to ask: Does the student evidence any remarkable individual differences or tendencies that make it challenging for him or her to be successful in the relevant contexts? The variables identified in this category often point us toward potential skill delays/deficits. For example, patterns of rigid thinking (a personal characteristic) may reflect weaknesses in behavioral flexibility (skill delay/deficit).

Step 7: Skill Delays/Deficits

The key here is to identify the particular skill delays and deficits associated with the interfering behavior. Skill delays/deficits may be related MOs. For example, delays in writing skills may be one of the reasons that a teacher's instruction to "get back to work" is aversive

and evokes interfering behaviors. Many of the students with whom we work present with numerous and pervasive behavior deficits, but you should list only the relevant ones.

More about Joey . . .

In the case of Joey, we also learned that he has diagnoses of intellectual disability and cerebral palsy, gets restless when his needs are not addressed immediately, gets bored easily when not engaged in specific activities, and generally has a hard time “being on his own, even for a short time” (personal characteristics). In addition, Joey presents with specific delays in expressive language and independent recreation-leisure skills (skill deficits/delays). Collectively, these personal characteristics and skill deficits help us make sense of the four-term contingency between MOs, S^{ps}, interfering behaviors, and reinforcing consequences.

Step 8: Parameters of Reinforcement

This can be a tough one. Most interfering behaviors are maintained by intermittent schedules of reinforcement, and consequences tend to be variable in terms of quality, magnitude, and timing. So unless you have really, really, really good and accurate data, you will only be able to estimate these parameters. The goal here is to capture how *robust* the reinforcing consequences are.

So, How Does This Apply to Joey’s Case?

With Joey, we summarized the parameters of reinforcement as follows:

- *Schedule*: Mom notices and responds to the behavior every 3–5 minutes (variable interval).
- *Quality*: High-value social interactions with Mom (soothing statements) that become even more valuable with deprivation (“absence makes the heart grow fonder”).
- *Magnitude*: Variable duration ranging from 30 seconds to 5 minutes of social attention.
- *Timing*: Variable but relatively immediate attention following interfering behavior.

CASE EXAMPLES OF THE BAPS MODEL

Now let’s put the model into action! The following are case examples designed to illustrate each component of the BAPS model.

The Case of the Missing Snack

Jeremy is a sixth-grade student who recently entered a middle school life skills program. Jeremy has diagnoses of intellectual disability and Prader–Willi syndrome. He also has a history of aggressive behaviors that have been reinforced by access to food items. For example, when his requests for food items are denied, he often engages in aggressive behaviors. Unfortunately, these incidents *sometimes* conclude with Jeremy gaining access to the restricted food items.

During the previous school year, a preference assessment was conducted, and results indicated strong preferences for various food items (e.g., salty snacks, candy bars, and baked

goods) and opportunities to swing (e.g., on the playground and in the occupational therapy room). A token reinforcement program was then implemented. Jeremy earned social reinforcement and tokens contingent on the *nonoccurrence* of aggressive behaviors during 15-minute intervals throughout the school day (differential reinforcement of other behavior; DRO) and the *occurrence* of target appropriate behaviors (differential reinforcement of alternative behavior; DRA). The tokens were exchanged for very small amounts of food items, access to the outdoor swing, or access to the indoor swing. The caloric value of snack items was restricted to very small amounts: approximately 150 calories per day. This intervention package resulted in a marked decrease in aggressive behaviors and an increase in appropriate behaviors, relative to baseline.

Over the summer, Jeremy participated in a recreation-leisure program where staff did not use the token reinforcement program. Jeremy was also evaluated at a regional medical clinic, and the team suggested intervening to address his weight gains over the past 3 years. Jeremy was subsequently placed on a restrictive diet (approximately 2,000 calories per day), which eliminated his access to unhealthy snack items (e.g., salty snacks, candy bars, and baked goods) and replaced them with healthy alternatives (e.g., celery chunks, cucumber slices, and radish bits).

Jeremy began the fall term at the middle school. Based on reports from the summer program staff, the middle school team decided to start the year without the token reinforcement program. It is also important to note that the middle school did not have either an outside swing or an indoor swing. No occurrences of aggressive behaviors were reported during the initial weeks of school. However, during the middle of the fourth week of the fall term, Jeremy engaged in aggressive behaviors toward a paraprofessional. It was reported that the aggression occurred “out of the blue.” Aggressive behaviors then began to occur more frequently: first two to three occurrences per week, then one to five occurrences per day. Moreover, aggressive behaviors began to generalize across people (e.g., paraprofessionals, teachers, peers, and community members), settings (e.g., transitions to and from the school bus, transitions within the school hallways, and community outings), and times of day (i.e., all school hours). The middle school team indicated that Jeremy was “getting out of control” and “becoming increasingly assaultive.” He was referred for an FBA.

The FBA was conducted by Fred Garvey (the school psychologist) and Ima Believer Jones (a predoctoral school psychology intern whose parents named all seven of their children after songs performed by the Monkees). The FBA included (1) record reviews; (2) review of previous behavioral interventions and data; (3) clinical interviews with parents, school staff, and Jeremy; and (4) descriptive assessments. Several observations were conducted during situations in which aggressive behaviors were likely to occur. The evaluators were able to observe three behavioral incidents, and they recorded relevant antecedents and consequences using the Incident-Based Functional Assessment Form (IBFAF). For each incident, they noted that Jeremy approached students or staff and attempted to either grab or open and “search” their backpacks. During interviews, they also learned that, on several occasions, Jeremy had taken backpacks from others and discovered high-calorie snack items. Although the team had attempted to restrict Jeremy’s access to backpacks, it was impossible to eliminate backpacks from a middle school environment or within community settings. The results of the FBA are summarized on the Behavior-Analytic Problem Solving: Assessment Results recording form (Figure 6.1).

Personal Characteristics: Traits, sensitivities, beliefs, and/or values contributing to interfering behavior • Preference for high-calorie snack foods (e.g., salty snacks, candy bars, and baked goods) • History of impulsivity	Context: Probable contexts in which interfering behavior occurs Transitions in hallway, arrival to and dismissal from school, community outings	Skill Delays/Deficits: Communication, academic, executive, adaptive living, social, and/or self-management skill delays/deficits contributing to interfering behavior • Functional communication skill delays • Self-management, self-control, and response inhibition weaknesses
Motivational “Triggers” (Motivating Operations): Deprivation of physical or social wants and needs, which increases the value of reinforcers and the probability of interfering behavior Presentation of aversive biological/medical conditions, emotional states, thoughts, academic tasks, or social situations, which increases the value of reinforcers and the probability of interfering behavior • Restricted diet (2,000 calories per day) increases the motivation for food • Deprivation of preferred high-calorie snack foods increases the motivation for unhealthy snacks • Restricted access to desired foods or locations where those foods may be obtained increases the motivation for access		
Sources of Reinforcement (Discriminative Stimuli): Persons, items, or locations that signal the availability of reinforcing consequences (i.e., source of reinforcement), thereby occasioning interfering behavior • Backpacks, lunchboxes, briefcases, and purses • Students, staff, and community members with backpacks, lunchboxes, briefcases, or purses		
Interfering Behavior(s): • Foraging behaviors: Opening unattended backpacks, lunchboxes, briefcases, purses, etc., and searching for snack items • Aggression: Hair pulling, pinching, slapping, and biting, directed toward peers, staff, and community members.		
Reinforcing Consequence(s): Consequence(s) that strengthen the future probability of the interfering behavior Access to preferred food items (individually mediated positive reinforcement)	Parameters of Reinforcement: Describe the power (robustness) of reinforcing consequences Schedule: • Intermittent access, approximately one out of every five attempts gains access to food Quality: • Access to highly preferred food items Magnitude: • Variable, ranging from a single bite of food to an entire chocolate bar Timing: • Immediate access to food items when attempts are successful	

FIGURE 6.1. Behavior-Analytic Problem Solving: Assessment Results recording form for Jeremy. Used with permission from the Margaret Murphy Centers for Children.

Additional Case Examples

In this section, we offer two additional case examples. For each case, the BAPS recording form was completed *following the completion of a comprehensive FBA*. We will not walk you through the details of the FBA process for these two cases, but we will highlight key details.

Figure 6.2 displays FBA results for Ricky, a student who displays loud vocalizations, verbal protests, and property destruction. The results of a comprehensive FBA revealed that these behaviors were reinforced by avoidance of the scheduled activity. Specifically, in response to interfering behaviors, the teacher verbally directed Ricky to the “quiet room.” Although this consequence was intended to function as a time-out from reinforcement to punish interfering behaviors, the intervention actually arranged socially mediated negative reinforcement, which served to strengthen the interfering behaviors. In this case, note that personal characteristics include rigidity, inflexibility, and resistance to change, and the corresponding skill delay/deficit pertained to a critical executive skill: flexibility. Also note that changes in task routines and schedules signaled a worsening condition (i.e., impending aversive circumstances) for Ricky and therefore functioned as MOs.

So, in this case . . .

- Do you see how the relationships among individual variables, antecedents, and consequences help us make sense of why Ricky exhibits interfering behavior?
- And, do you see how identifying these variables could lead to an effective multicomponent function-based intervention plan (e.g., eliminating the time-out procedure, reinforcing task participation, teaching behavioral flexibility and strategies for coping with change, modifying the context to minimize changes in expectations and routines)?

Figure 6.3 summarizes our final case, which involves a student, Maddie, who engages in disruptive interfering behaviors such as swearing, making profane gestures, stomping her feet, pushing furniture, and slamming doors. This is a complex case. Results of a comprehensive FBA revealed that Maddie’s interfering behaviors are members of the same response class and typically occur as a cluster. These behaviors are reinforced both by socially mediated negative reinforcement (i.e., avoidance or cessation of difficult academic tasks) and automatic negative reinforcement (i.e., reduction of anxiety and negative self-talk). Note that *anxiety* (described by Maddie as “feeling tense, tied up in knots, and ready to explode”) occurs primarily within the context of math instruction. Take a look at the contributing personal characteristics, skill delays/deficits, motivational triggers, and sources of reinforcement. Can you see how the variables in this case collude to evoke and maintain Maddie’s interfering behavior?

COMMUNICATING FBA RESULTS AND RECOMMENDATIONS USING THE BAPS MODEL

We recommend that you include the completed Behavior-Analytic Problem Solving: Assessment Results recording form as an appendix to your evaluation report. And when FBAs address multiple interfering behaviors, we recommend including multiple recording forms

Personal Characteristics: Traits, sensitivities, beliefs, and/or values contributing to interfering behavior • Rigid thinking • Resistant to change	Context: Probable contexts in which interfering behavior occurs Situations that unexpectedly deviate from the regular or expected routine	Skill Delays/Deficits: Communication, academic, executive, adaptive living, social, and/or self-management skill delays/deficits contributing to interfering behavior Weaknesses in flexibility (the ability to adapt to changing conditions and revise plans in the face of obstacles, setbacks, new information, etc.)
Motivational “Triggers” (Motivating Operations): Deprivation of physical or social wants and needs, which increases the value of reinforcers and the probability of interfering behavior Presentation of aversive biological/medical conditions, emotional states, thoughts, academic tasks, or social situations, which increases the value of reinforcers and the probability of interfering behavior Presentation of unexpected changes in tasks or schedules		
Sources of Reinforcement (Discriminative Stimuli): Persons, items, or locations that signal the availability of reinforcing consequences (i.e., source of reinforcement), thereby occasioning interfering behavior • Staff who have implemented the “time-out” procedure • Availability of the “quiet room”		
Interfering Behavior(s): • Loud vocalizations (screaming) • Vocal protest behavior (saying “no” or “I don’t want to _____”) • Property destruction (ripping up materials or overturning furniture)		
Reinforcing Consequence(s): Consequence(s) that strengthen the future probability of the interfering behavior Avoidance of the unexpected/changed task or activity (socially mediated negative reinforcement)	Parameters of Reinforcement: Describe the power (robustness) of reinforcing consequences Schedule: • Continuous for property destruction (i.e., property destruction always results in “time-out”) Quality: • High (avoidance and access to a break with staff support) Magnitude: • Variable, time-out for 5–10 minutes Timing: • Immediate redirection to time-out	

FIGURE 6.2. Behavior-Analytic Problem Solving: Assessment Results recording form for Ricky. Used with permission from the Margaret Murphy Centers for Children.

Personal Characteristics: Traits, sensitivities, beliefs, and/or values contributing to interfering behavior	Context: Probable contexts in which interfering behavior occurs	Skill Delays/Deficits: Communication, academic, executive, adaptive living, social, and/or self-management skill delays/deficits contributing to interfering behavior
• Sensitive to corrective feedback from teachers • Perfectionism • Quick to anger	Math instruction (algebra class)	• Weaknesses in math skills • Weaknesses in self-management/emotional regulation • Weaknesses in psychological flexibility

Motivational “Triggers” (Motivating Operations): Deprivation of physical or social wants and needs, which increases the value of reinforcers and the probability of interfering behavior Presentation of aversive biological/medical conditions, emotional states, thoughts, academic tasks, or social situations, which increases the value of reinforcers and the probability of interfering behavior
• Presentation of challenging math instruction and corrective feedback increases the value of escape • Aversive internal arousal (anxiety) increases the value of arousal reduction (feeling calm again) • Irrational self-statements (e.g., “I can’t do anything right,” “I suck at math”) increase the value of escaping math and “turning off” those thoughts

Sources of Reinforcement (Discriminative Stimuli): Persons, items, or locations that signal the availability of reinforcing consequences (i.e., source of reinforcement), thereby occasioning interfering behavior
• Teacher with a history of backing off and/or sending disruptive students out of the classroom • Presence of doors and chairs to be shoved and slammed

Interfering Behavior(s):
Disruptive behaviors: Screaming, swearing, making profane gestures, stomping feet, shoving chairs, and slamming doors

Reinforcing Consequence(s): Consequence(s) that strengthen the future probability of the interfering behavior	Parameters of Reinforcement: Describe the power (robustness) of reinforcing consequences
• Escape from challenging math expectations (socially mediated negative reinforcement) • Termination of corrective feedback (socially mediated negative reinforcement) • Reductions in arousal/anxiety levels (automatic negative reinforcement) • Reductions in negative self-statements (automatic negative reinforcement)	Schedule: • Intermittent social reinforcement, continuous automatic reinforcement Quality: • High-quality escape from math in the guidance counselor’s office Magnitude: • Variable, ranging from 15–30 minutes out of class Timing: • Immediate

FIGURE 6.3. Behavior-Analytic Problem Solving: Assessment Results recording form for Maddie. Used with permission from the Margaret Murphy Centers for Children.

in the report. The BAPS recording form provides a concise synthesis of FBA results and summarizes essential information in a visual format that facilitates the communication of results to caregivers and school teams. It is our standard practice to meet with parents/guardians to review and discuss evaluation reports prior to individual education program (IEP) team meetings, and we use the BAPS recording forms to guide these discussions. Finally, we find that a comprehensive review of the BAPS recording form is invaluable when collaborating with teams to identify and design individualized, function-based interventions. (*Note:* Chapter 11 provides additional information about using FBA data to drive intervention planning and includes a complementary form—Behavior-Analytic Problem Solving: Function-Focused Intervention—to support this process.) The following subsection on “revelations” provides examples of the type of feedback we receive when we present FBA findings using the Behavior-Analytic Problem Solving: Assessment Results recording form and thoroughly explain how each component in the BAPS model contributes to occurrences of interfering behavior.

BAPS Revelations

Reviewing the completed Behavior-Analytic Problem Solving: Assessment Results recording form in the context of collaborative problem-solving meetings with members of school teams (especially those trained in the conceptual foundations of the BAPS model) facilitates a comprehensive understanding of the variables that trigger and maintain interfering behaviors. In our experience, when evaluators use the recording form as a visual aid to support these discussions, several revelations occur:

The “Aha” Moment

This is when team members report, “This makes sense. I get it now. Sally’s behavior is not ‘out of the blue.’ Her behavior is predictable and lawful.”

The Perfect Storm

Now team members offer the reflection “Holy cow! Given (1) the school context, (2) Kristina’s personal characteristics, strengths, and weaknesses, (3) motivational triggers, (4) sources of reinforcement, and (5) reinforcing contingencies, it’s no wonder she behaves that way!”

The Unfortunate Realization about the Ideal Environment for Shaping Interfering Behavior

This is when a team member exclaims with awe and wonder, “This illustrates the perfect setup for teaching: (1) establishing high motivation, (2) arranging multiple and varied stimulus conditions for learning, (3) offering repeated opportunities to practice the behavior, and (4) delivering robust reinforcement. OMG! We have inadvertently been teaching Johnny to be disruptive! We need to change tactics . . . and quickly!”

The Student Perspective

The BAPS model helps us understand interfering behaviors from the student's point of view. Perhaps the student would say, "Duhhh. Of course I act this way. I'm the victim of an environment that is engineered to shape disruptive behavior . . . it's so unfair."

The Time for Advocacy

In the end, we hope that team members say, "We need to drop the blame game. We've got to quit blaming Peggy. Let's quit talking about Peggy and *her* problems. Let's get off the diagnosis merry-go-round . . . it has not helped us so far. Obviously these behaviors are not her fault. She didn't *choose* to act this way. The interfering behavior is the result of a dynamic interaction between her unique characteristics and the environment that we created. We're *all* part of the problem, so we *all* need to be part of the solution. This BAPS model provides our team with a basis for designing a Peggy-centered function-based treatment plan, so it's time we act on her behalf and advocate for a comprehensive multifaceted intervention. To quote the learned advice of the sage and philosopher Bluto [portrayed by John Belushi in the film *Animal House*] when faced with a time for decisive action: 'Let's just do it!'"

Summary

So yeah, "let's just do it." Armed with an understanding of the variables that evoke, occasion, and reinforce interfering behaviors, we can work effectively with school teams to collaboratively design interventions. Most importantly, when we use the BAPS recording form to review FBA results with members of the problem-solving team, several things happen:

- *People get it.* They understand why a student engages in interfering behavior.
- *Interventions become self-evident.* When team members understand the function of interfering behaviors, appropriately matched interventions become obvious.
- *Team members "buy in."* When team members understand why interfering behaviors occur and the underlying rationale for function-based interventions, intervention acceptability ratings soar and team members proceed with increased confidence in the intervention plans.
- *Team members follow through.* When team members understand both why and how to intervene effectively, they tend to do so with consistency and accuracy, which is the key to achieving treatment integrity.

SUMMARY

So there you have it. Wow! We've covered a lot here. Make no mistake, the process of synthesizing and interpreting data obtained from a comprehensive FBA can be arduous. But, it is also an energizing and fulfilling endeavor. We hope that you find the BAPS recording form to be a valuable tool as you uncover the mystery of student behavior. And remember, have fun!

FORM 6.1

Behavior-Analytic Problem Solving: Assessment Results

Name: _____ Date: _____

Personal Characteristics: Traits, sensitivities, beliefs, and/or values contributing to interfering behavior	Context: Probable contexts in which interfering behavior occurs	Skill Delays/Deficits: Communication, academic, executive, adaptive living, social, and/or self-management skill delays/deficits contributing to interfering behavior

Motivational “Triggers” (Motivating Operations): Deprivation of physical or social wants and needs, which increases the value of reinforcers and the probability of interfering behavior Presentation of aversive biological/medical conditions, emotional states, thoughts, academic tasks, or social situations, which increases the value of reinforcers and the probability of interfering behavior

Sources of Reinforcement (Discriminative Stimuli): Persons, items, or locations that signal the availability of reinforcing consequences (i.e., source of reinforcement), thereby occasioning interfering behavior

Interfering Behavior(s):

Reinforcing Consequence(s): Consequence(s) that strengthen the future probability of the interfering behavior	Parameters of Reinforcement: Describe the power (robustness) of reinforcing consequences
	Schedule: Quality: Magnitude: Timing:

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